

The Danish N-Footprint: The Cost of Consumption and the N_r Conundrum Beyond Borders

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KEY CLAIM

Denmark shows that efficient agricultural production does not guarantee a small N footprint or N independence when domestic consumption, export demand and imported N_r sustain large cross-border N flows.

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Denmark's N footprint was estimated at 27.5 kg N capita⁻¹ yr⁻¹, of which food production and consumption accounted for 22.7 kg N capita⁻¹ yr⁻¹, or 82%. Meat and meat-derived products contributed 71% of the food N footprint. Denmark had the fourth-highest footprint among the 11 national estimates compared, while its mean food virtual nitrogen factor (VNF) was the third lowest among 12 country calculations. These findings reveal the central Danish N conundrum: comparatively efficient production does not necessarily produce a small footprint when consumption is high, and animal products dominate the footprint.

The longitudinal pork-system account demonstrates how this relationship develops through agricultural production and trade. Between 1995 and 2023, whole-chain nitrogen-use efficiency (NUE) increased from 16.7% to 29.2%, while the non-recovery VNF declined from 4.99 to 2.42 kg N_r kg⁻¹ pork-product N. Nevertheless, total pork-product N increased by 35.4%, and output growth absorbed 47.2% of the potential input savings generated by higher efficiency.

Export demand was substantially larger than domestic pork demand. In 2023, the border account recorded 48.95 million kg N in pork-related exports compared with 7.45 million kg N in domestic pork consumption. Live exports increased to 30.2% of total pork-product N, shifting the remaining finishing-stage feed requirements, manure production and associated N burden to importing countries. At the same time, national gross synthetic-fertilizer N_r imports increased by 37.8%.

The remaining challenge is therefore interpretive. Demand determines the scale of production, efficiency determines the N required per unit of output, and trade determines where N inputs originate and where the remaining production burden occurs. These three dimensions must be assessed together.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

This contribution connects consumption-based N footprints with production-based NUE, VNF and mass-balance accounts. N footprints identify the demand creating N pressure; NUE and VNF describe how efficiently that demand is supplied; and trade-linked mass balances reveal where N enters, moves through and leaves the agricultural system. Combining these indicators helps distinguish genuine production improvements from changes caused by expanding output or relocating production stages across borders.

QUESTIONS TO SEED THE DISCUSSION

1. What minimum combination of N footprint, NUE, VNF and trade indicators is needed to prevent high efficiency from being interpreted as low absolute N pressure?
2. Should dependence on imported N_r be reported alongside NUE and VNF when assessing national food-system performance?

REFERENCES

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